ASSIGNMENT - WORKSTATION IMAGE EXAMINATION I - WINHEX

OVERVIEW

Some corporate documents were stolen from an executive's workstation and posted on a competitor's website. Your firm has been hired to perform the analysis. A forensic image of the workstation has been provided.

INSTRUCTIONS

> Case Number: YY-T005

A forensic analyst must be familiar with using a hexadecimal editor. For this assignment we will be using WinHex or 010 Editor to review our recently acquired USB image and answer questions regarding disk partition layouts. For this assignment you will need your Windows forensic virtual machine and a hexadecimal editor.

1. Download and unzip the image file
2. Verify the image
3. Open the file in WinHex or 010 Editor
4. Navigate to offset 0 and answer the following questions

QUESTIONS

Review the first 446 bytes

1. Does this image have boot code in the first 440 bytes?
2. Give your observations including any plaintext in the code.
3. What is the "disk signature" starting at offset 0x01B8?
    - Provide the value as a hexadecimal value like 0x01234567
    - Give the correct value with endianness in consideration

Review the partition table beginning at offset 0x01BE

1. How many partition entries are in the table? Include empty ones.
2. Which partition entry is marked bootable?
3. Partition entry 1 - What is the partition entry type?
    - Give the hexadecimal value and the type (e.g. Linux, FAT32)
4. Partition entry 1 - What is the LBA start address?
    - Give the hexadecimal value with endianness in consideration
5. Calculate the offset of the LBA start address.
    - Sector size is 512 bytes or 0x200 in hexadecimal.
6. Partition entry 1 - What is the partition size?
    - Give the hexadecimal value with endianness in consideration
7. Calculate the size of the partition in bytes.
    - Sector size is 512 bytes or 0x200 in hexadecimal
    - Size must be in decimal
8. What is the difference, in bytes, between the actual size of the evidence file versus the value above?

Navigate to Partition Entry One LBA Start Address Offset

1. What are the first 8 bytes at this offset?
    - DO NOT consider endianness in this value
2. Does this value support your partition type answer above?
3. Give your observations including any plaintext in the code.

4. Based on your observations, does this disk image use an MBR or GPT partitioning scheme? Explain why.

EVIDENCE FILES

Filename	windows-image-01.dd
Type	DOS/MBR boot sector MS-MBR XP english at offset 0x12c “Invalid partition table” at offset 0x144 “Error loading operating system” at offset 0x163 “Missing operating system”, disk signature 0x39bf39be; partition 1 : ID=0x7, active, start-CHS (0x0,1,1), end-CHS (0x3ff,254,63), startsector 63, 20948697 sectors
Size	10737418240 bytes
MD5	78a52b5bac78f4e711607707ac0e3f93
SHA1	ba7dc57e08bb6e3393aee15c713ae04feadcd181
SHA256	4d4fc46f284630a69980340ee36d3ed486de424a9f456eedba0f7194cc7a991b

REQUIRED TOOLS

Name	Vendor
WinHex	xWays
010 Editor	SweetScape

REQUIRED FILES

- evidence-report-template.docx

DELIVERABLE REQUIREMENTS

- Evidence report with the following sections and requirements
  - Executive Summary
  - Evidence Seized
  - Tools Used
  - Case Information, Observables, Investigator Processes
    - Question answers
  - Conclusion
  - Exhibits (Not required)
  - References (Not required)
  - Photos/screenshots, w/ captions, are required to justify your findings
    - In line with your question answers, not in an appendix/exhibit
  - Must use report template

GRADING RUBRIC

- Evidence report - 100 pts
  - Completed answers - 60 pts
    - Answered questions for each item - 2 pts
    - Provided screenshot for each item - 2 pts
  - Used template - 30 pts
  - Following instructions - 10 pts